

Credit Risk Prediction: A Data-Driven Approach to Loan Default Classification

Introduction

Credit default represents a major source of financial risk for retail lending, contributing directly to financial loss. Recent industry data indicates that the proportion of consumers 90 or more days past due increased from 2.34% in Q3 2023 to 2.43% in Q3 2024, reaffirming the growing importance of robust credit risk assessment frameworks. Equally, the EU AI Act categorises AI systems used to evaluate creditworthiness as high-risk applications. Therefore, ensuring greater transparency and explainability in how these systems reach decisions has become a core regulatory expectation for lenders. With lending volumes increasing, institutions are relying more heavily on data-driven models to refine credit decisioning.

This study examines how effectively applicant financial and credit-related characteristics can predict loan default and identifies the features most indicative of repayment failure. To address this, an end-to-end data science workflow is implemented, incorporating data preprocessing, exploratory analysis, feature engineering, and supervised classification modelling. The analysis evaluates the predictive performance of multiple classification approaches, highlighting variables that contribute most strongly to model accuracy. In doing so, the study demonstrates how machine learning can support transparent, evidence-based credit decisioning.

Focused Analysis

Data Understanding

This study uses the Credit Risk Dataset from Kaggle which contains applicant financial, employment, and credit-related attributes used to predict loan default. The dataset comprises 32,581 observations across 12 features, with *Loan_status* serving as the binary target variable. These variables collectively provide insight into applicant affordability, creditworthiness, and repayment behaviour, forming a suitable basis for supervised classification.

Missing values, outliers, and inconsistent entries were present across both numerical and categorical fields, including unrealistic applicant ages and incomplete employment-length records. Addressing these issues was essential to improve data reliability and ensure that subsequent modelling stages were grounded in a robust and internally consistent dataset.

```
person_age: 0 missing values
person_income: 0 missing values
person_home_ownership: 0 missing values
person_emp_length: 895 missing values
loan_intent: 0 missing values
loan_grade: 0 missing values
loan_amnt: 0 missing values
loan_int_rate: 3116 missing values
loan_status: 0 missing values
loan_percent_income: 0 missing values
cb_person_default_on_file: 0 missing values
cb_person_cred_hist_length: 0 missing values

credit_df['person_age'].unique()
✓ 0.0s

array([ 22,  21,  25,  23,  24,  26, 144, 123,  20,  32,  34,  29,  33,
        28,  35,  31,  27,  30,  36,  40,  50,  45,  37,  39,  44,  43,
        41,  46,  38,  47,  42,  48,  49,  58,  65,  51,  53,  66,  61,
        54,  57,  59,  62,  60,  55,  52,  64,  70,  78,  69,  56,  73,
        63,  94,  80,  84,  76,  67])
```

Figure 1: Identification of null values, erroneous entries requiring preprocessing

Data Analysis

Data cleaning involved removing 165 duplicate rows, eliminating erroneous values in applicant age and employment length, dropping null values in interest rate and employment length each accounting for less than 10% of the dataset, and capping income at the 99th percentile to mitigate right skew. Exploratory analysis was conducted to identify patterns relevant to default risk. Class distribution analysis showed a 78/22 split between non-defaulting and defaulting applicants, confirming class imbalance and highlighting the need for metrics beyond accuracy.

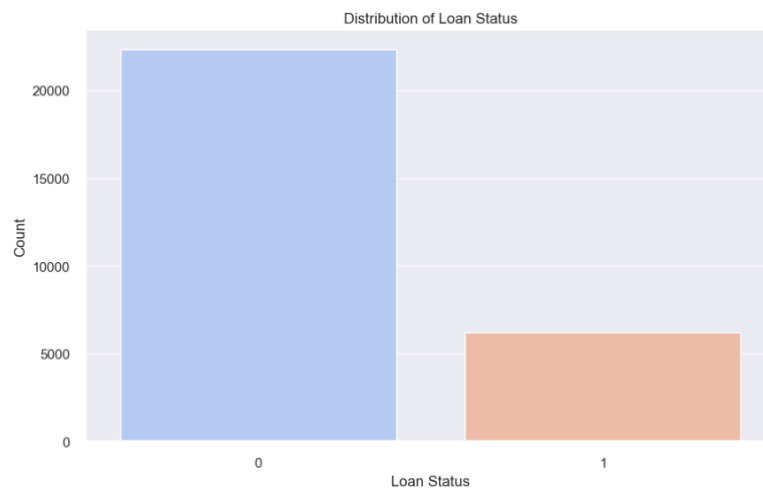


Figure 2: Class Distribution Shown with loan_status Countplot

Univariate distributions revealed a right skew across financial features. Bivariate relationships between continuous features and loan status were investigated using box plots, chosen for their ability to compare distributional spread and median across target groups simultaneously. This revealed that defaulters consistently carried higher loan-to-income ratios and interest rates. A violin plot further confirmed defaulters are disproportionately concentrated at higher affordability ratios.

Applicant Characteristics vs Loan Default

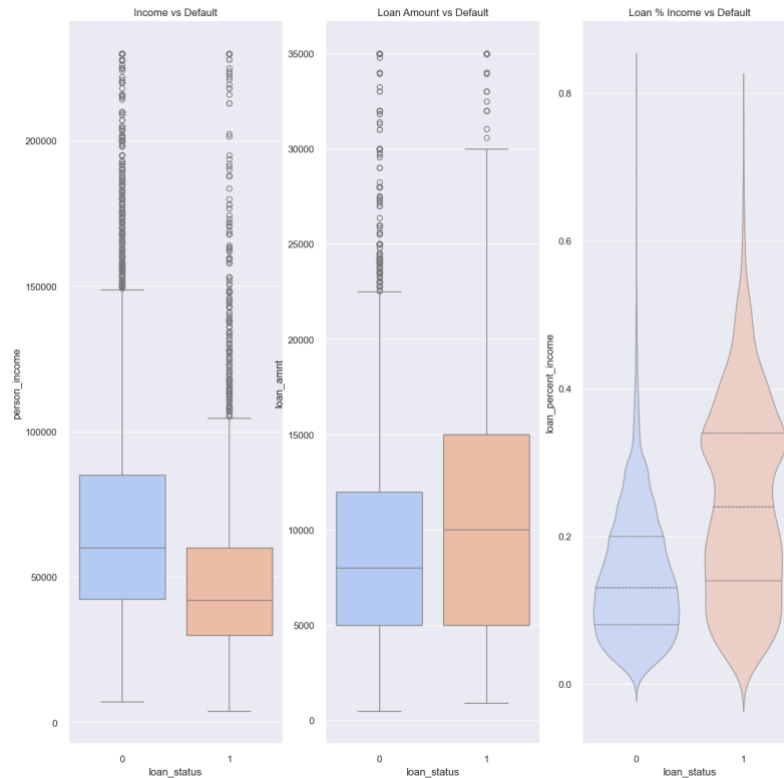


Figure 3: Bivariate Financial Feature Analysis with Box Plots and Violin Plot

For categorical features, default rates were computed using grouped aggregation visualised as bar charts. Loan grade displayed a near-monotonic relationship with default probability, ranging from under 10% for Grade A to almost 98% for Grade G. Applicants with a prior default on file defaulted at roughly twice the rate of those without. A correlation heatmap confirmed loan-to-income ratio ($r=0.38$) and interest rate ($r=0.34$) as the strongest linear associations with loan status, whilst a correlation of 0.88 between age and credit history length indicated partial feature redundancy.

Categorical Credit Indicators vs Loan Default

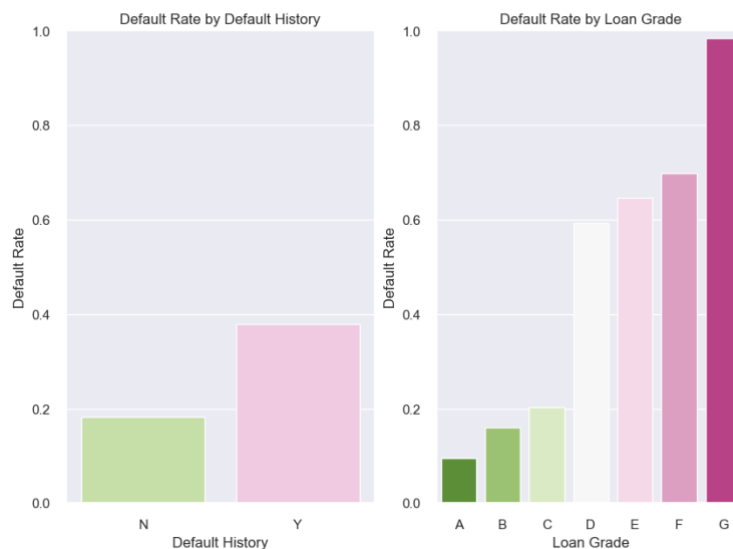


Figure 4: Categorical Default Rates

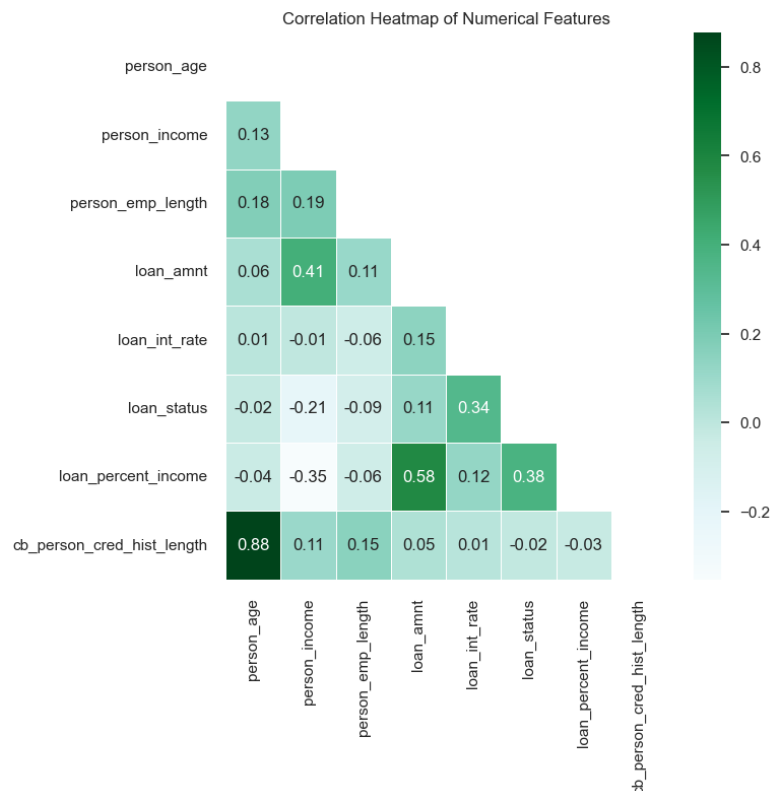


Figure 5: Correlation Heatmap

Modelling

Three supervised classification models were implemented across linear and ensemble-based learning paradigms: Logistic Regression as an interpretable baseline, Random Forest to capture non-linear feature interactions through parallel ensemble learning, and XGBoost for its superior performance on structured tabular data through sequential gradient boosting.

Categorical features were encoded using binary mapping and one-hot encoding prior to modelling. Given the class imbalance in target distribution, stratified train-test splitting and cross-validation were applied to preserve class proportions during training and evaluation. For XGBoost, the `scale_pos_weight` was set using the ratio between defaults and non-defaults class observations, improving sensitivity towards the minority class. Hyperparameter optimisation was conducted via GridSearchCV with stratified k-fold cross-validation to improve generalisation by evaluating performance across resampled splits.

Model performance was assessed using precision, recall, F1 score and ROC-AUC. Accuracy was not used as a primary metric due to the skewed class distribution, as it would overstate performance by favouring the majority class. Recall and F1 score were prioritised because false negatives in credit risk prediction represent a real business risk, with high-risk borrowers potentially being incorrectly approved. Comparative evaluation showed that XGBoost achieved the strongest overall performance, offering the best balance between minority-class recall and overall discriminative ability.

	Model	Accuracy	Precision (Class 1)	Recall (Class 1)	F1 (Class 1)	ROC-AUC
0	Logistic Regression	0.82	0.55	0.78	0.65	0.8700
1	Random Forest	0.93	0.97	0.72	0.83	0.9340
2	XGBoost (Tuned)	0.92	0.81	0.80	0.80	0.9468

Figure 6: Model Comparison Results Table

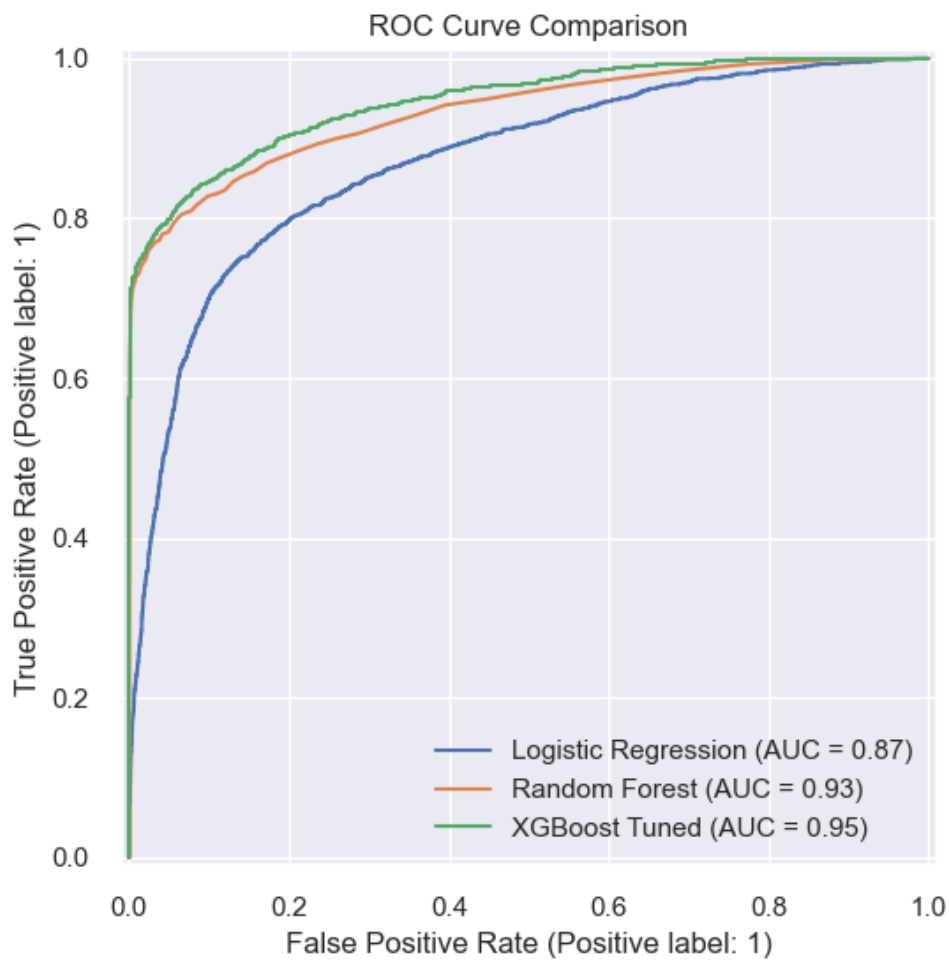


Figure 7: Comparative ROC AUC Curve

Insight Summary

Feature importance identified loan-to-income ratio as the strongest predictor of default, consistent with EDA findings showing defaulters clustered at higher affordability ratios. Interest rate and loan grades emerged as the next influential features, reinforcing that lender-assigned risk indicators carry genuine predictive signal. Home ownership status, also appeared prominently among the top features, suggesting that housing instability is a meaningful indicator for broader financial vulnerability.

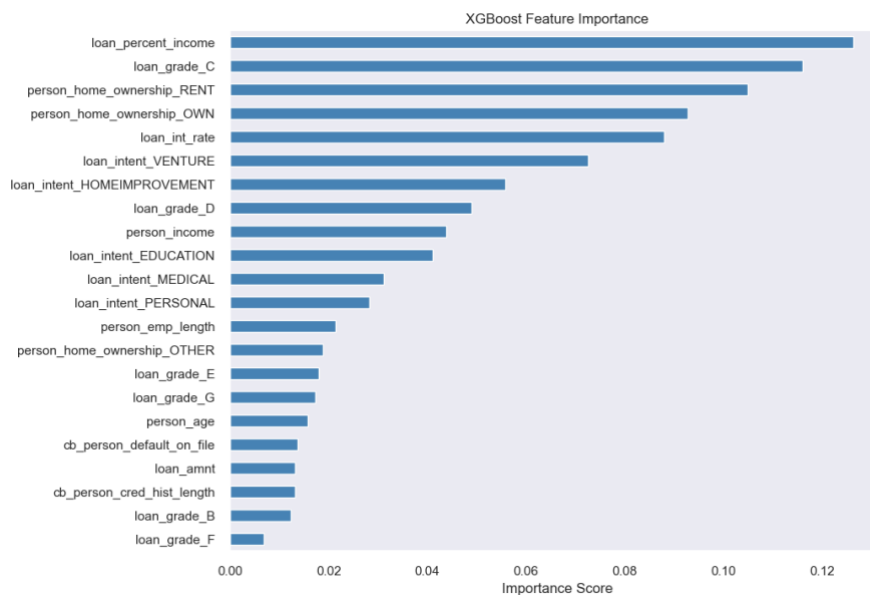


Figure 7: Feature Importance Chart

These findings provide several actionable recommendations for a retail lending company. Firstly, loan-to-income ratio should serve as a primary screening criterion, with applicants borrowing above 40% of their income being flagged for additional review. Secondly, venture-purpose loan applications should include further income verification given their higher rate of default. Third, whilst the XGBoost model achieved a ROC-AUC of 0.9468 and 92% accuracy, a recall of 80% on the default class means approximately one in five defaulters would still be approved – showing the model should accompany human credit judgement as opposed to replacing it in alignment with EU AI Act’s transparency requirements.

Conclusion

This study faced two main challenges. The first was class imbalance, with defaulters making up only 22 percent of the dataset. This was managed through stratified splitting, class-weighted logistic regression and the `scale_pos_weight` parameter in XGBoost, which improved recall for the minority class. The second challenge involved erroneous outliers in applicant age and employment length, which were identified using IQR-based detection and removed to maintain data quality. A limitation of the approach is that synthetic oversampling methods such as SMOTE were not applied. Future work would test SMOTE to assess whether it can further improve default recall.

Reference:

TransUnion (2024) *Balances and Delinquencies Begin to Stabilize Across Many Credit Products: Q3 2024 Quarterly Credit Industry Insights Report*. TransUnion. Available at: <https://newsroom.transunion.com/q3-2024-ciir/> (Accessed: 10 May 2026).

European Parliament and Council of the European Union (2024) *Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act)*. Official Journal of the European Union. Available at: <https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai> (Accessed: 10 May 2026).

Laoste (2021) *Credit Risk Dataset*. Kaggle. Available at: <https://www.kaggle.com/datasets/laotse/credit-risk-dataset> (Accessed: 30 April 2026).